Project 2 Report: Employee Attrition

Stacy Bowler
Utah State University
stacy.bowler@sdl.usu.edu

Derrik Miller
Utah State University
derrik.miller@sdl.usu.edu

2026-08-07

Abstract This project develops a predictive employee attrition model using regularized logistic regression. The objective is to identify employees at elevated risk of leaving and determine whether targeted retention interventions provide measurable financial value. Multiple modeling approaches were evaluated using cross-validation, predictive performance metrics, and a business utility function incorporating intervention costs and retention benefits. An Elastic Net logistic regression model was selected based on its balance of predictive accuracy, precision, and expected financial utility. The final model generated positive expected value on held-out test data, demonstrating the potential for data-driven retention strategies.

Introduction

The objective of this project is to develop a predictive model capable of determining which employees are most likely to leave. This can allow the business to focus their retention efforts on high-risk employees.

The business decision addressed by this analysis is:

> How much financial value does using this predictive model create compared with not using one?

The resulting model can be used to identify employees that appear likely to leave.

Business Objective

The primary objective is to accurately determine which employees are likely to quit before they do.

A practical utility function for this problem is:

$$U = \begin{cases} 0.60 \times 40000 - 8000 = 16000 & \text{if a retention intervention is provided and prevents attrition} \\ -8000 & \text{if a retention intervention is provided unnecessarily} \\ 0 & \text{if no intervention is provided and the employee leaves} \\ 0 & \text{if no intervention is provided and the employee stays} \end{cases}$$

The utility function reflects the incremental financial impact of acting on the model's recommendations. A true positive represents the expected net savings from successfully retaining an employee, while a false positive represents the cost of an unnecessary retention intervention. False negatives and true negatives are assigned zero utility because they represent the baseline outcome in which no intervention occurs.

The utility function is based on the following assumptions:

- Cost of employee turnover: $40,000
- Cost of retention intervention: $8,000
- Retention intervention success rate: 60%

This utility framework prioritizes finding employees where intervention is likely to create value. False positives have a measurable cost because they represent unnecessary retention efforts, while false negatives represent missed opportunities to intervene.

Plan

The analysis follows the modeling workflow:

1. Define the business problem.
2. Acquire and clean employee data.
3. Explore relationships between candidate predictors and attrition.
4. Build an interpretable regression model.
5. Evaluate assumptions and model fit.
6. Generate predictions and business recommendations.

Important predictors expected to influence employee attrition included:

- Age
- Attrition
- Compensation
- Years at the company
- Overtime status
- Job role
- Job satisfication score

The available dataset contained these factors along with additional demographic, job, and employment history variables, providing a broad set of candidate predictors.

Build

Data Source

The dataset was obtained from Kaggle at https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset and contains a fictional dataset created by IBM data scientists.

The dataset initially contained 35 predictors, including:

- Age
- Attrition
- Department
- Gender
- Job Role
- Job Satisfaction
- Marital Status
- Monthly Income
- Number of Companies Worked
- Overtime
- Years Since Last Promotion
- Years at Company
- Years in Current Role

Data Cleaning

The following preprocessing steps were performed:

- Converted Attrition to a binary outcome where Yes = 1

```
import os
import numpy as np
import polars as pl
import pandas as pd
import seaborn.objects as so
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.base import clone
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import (
  confusion_matrix,
  accuracy_score,
  precision_score,
  recall_score,
```

```
  roc_auc_score
)

pl.Config.set_tbl_rows(-1)
pl.Config.set_tbl_cols(-1)
pl.Config.set_fmt_str_lengths(100)

employee_data = (pl.read_csv(os.path.join('data', 'HR-Employee-Attrition.csv'))
  # Make the outcome binary where 'Yes' = 1
  .with_columns(
    pl.when(pl.col('Attrition') == 'Yes').then(1)
    .when(pl.col('Attrition') == 'No').then(0)
    .alias('Left')
  )
        .select(pl.exclude(['Attrition',    'Department',    'EmployeeNumber',
'DailyRate', 'MonthlyIncome', 'EmployeeCount', 'Over18', 'PerformanceRating',
'StandardHours']))
)
```

Explore

Exploratory Data Analysis

```
(
  employee_data
  .group_by("Left")
  .agg(count=pl.len())
)
```

Left	count
i32	u32
1	237
0	1233

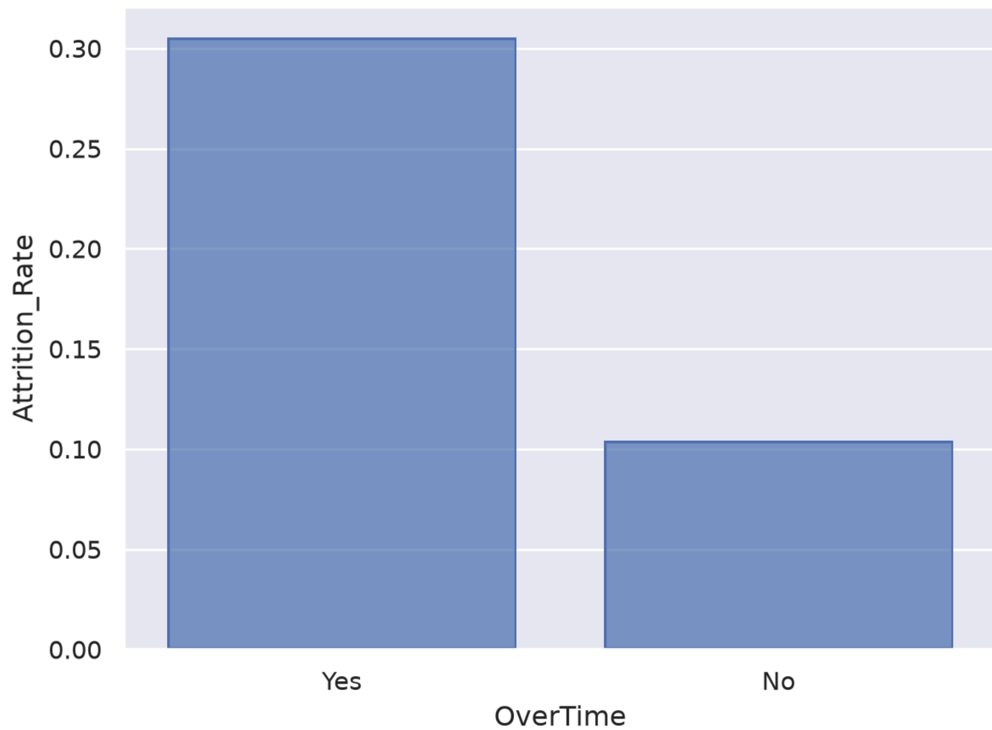
The dataset contains a relatively small proportion of employees who leave, making accuracy alone an insufficient evaluation metric. A model that predicted every employee would stay would achieve high accuracy but provide no business value. Therefore, this analysis prioritizes utility, recall, and precision in addition to accuracy.

Visualizations

Initial visualizations focused on the relationships between attrition and major numerical predictors.

Overtime vs Attrition

```
overtime_rate = (
  employee_data
  .group_by("OverTime")
  .agg(Attrition_Rate=pl.col("Left").mean())
)

so.Plot(overtime_rate.to_pandas(),
    x="OverTime",
    y="Attrition_Rate") \
  .add(so.Bar())
```

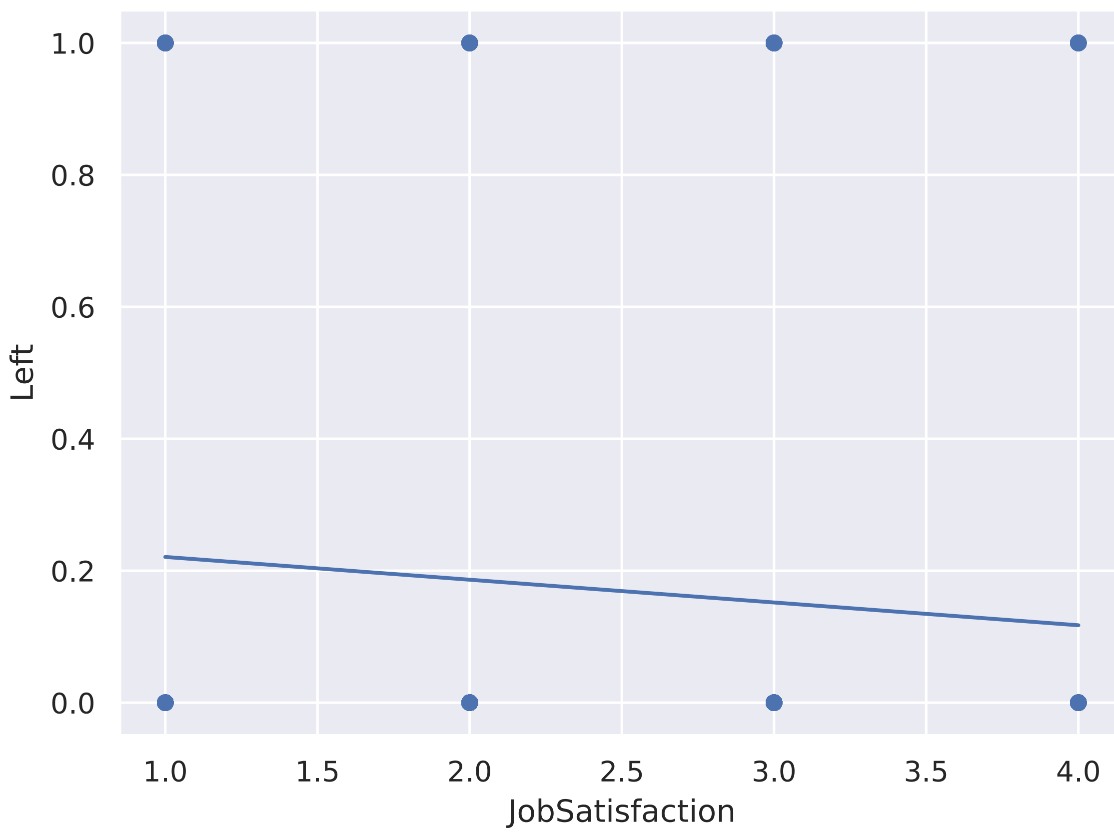


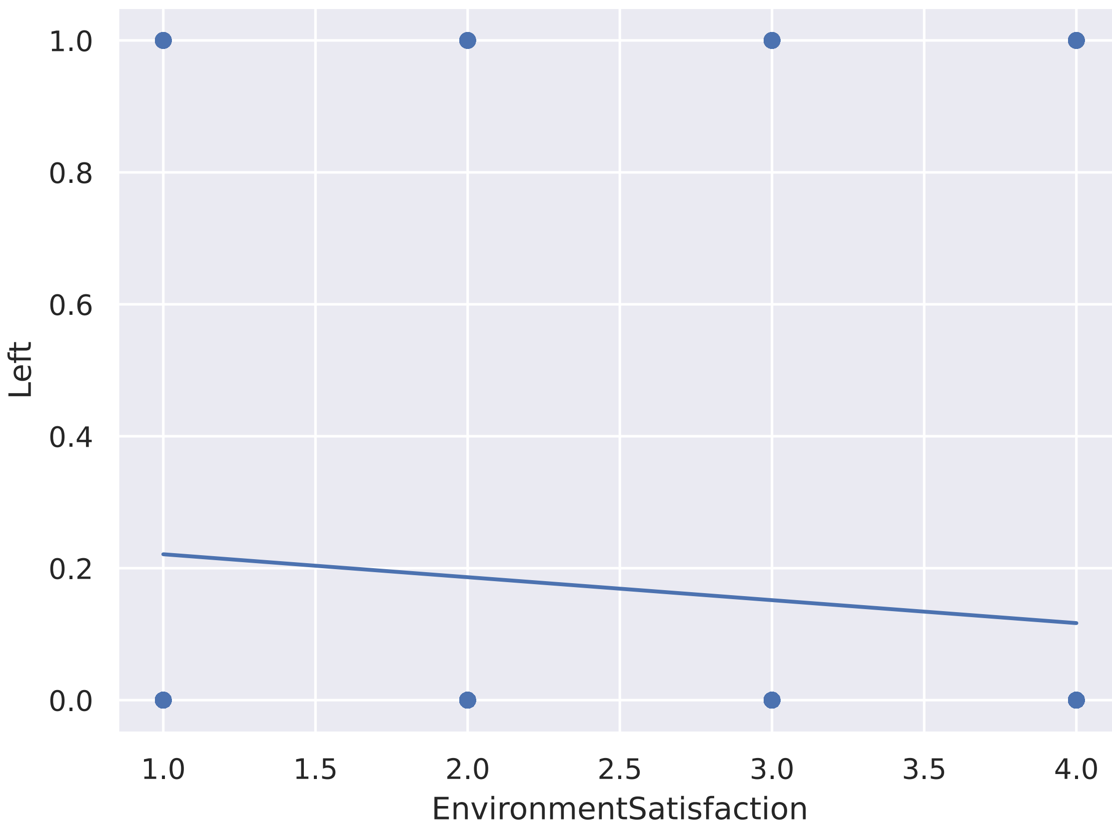
This looks like employees that work overtime are more likely to leave.

Satisfaction vs Attrition

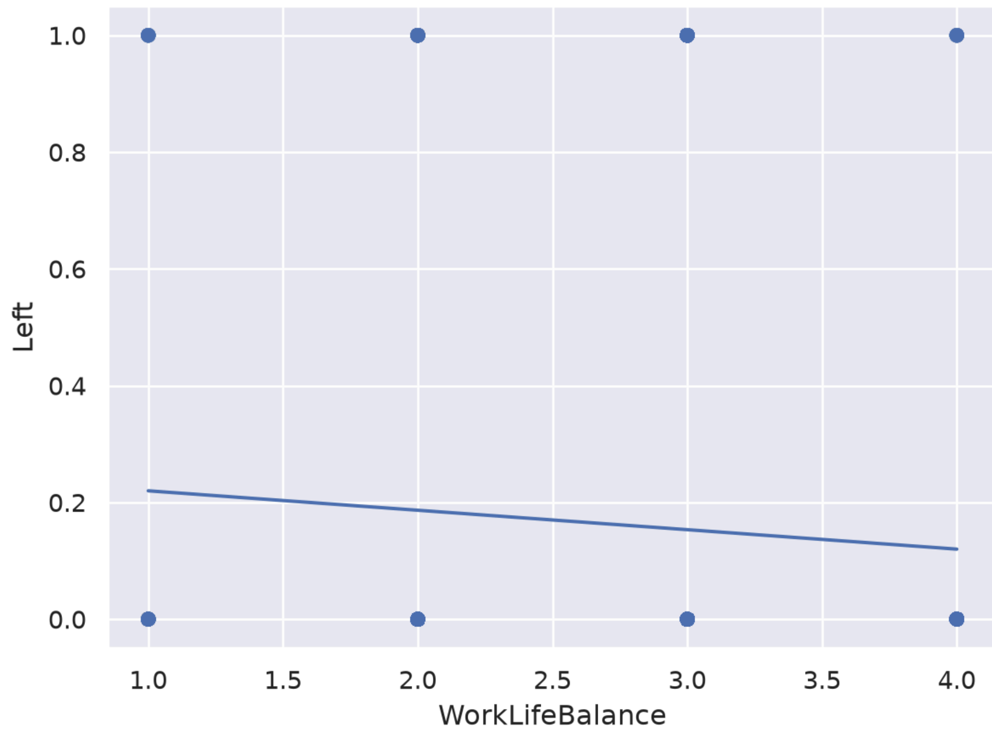
```
(
  so.Plot(employee_data.to_pandas(),
          x="JobSatisfaction",
          y="Left")
  .add(so.Dot(alpha=.20))
  .add(so.Line(), so.PolyFit(order=1))
```

```
).show()

(
  so.Plot(employee_data.to_pandas(),
          x="EnvironmentSatisfaction",
          y="Left")
  .add(so.Dot(alpha=.20))
  .add(so.Line(), so.PolyFit(order=1))
).show()

(
  so.Plot(employee_data.to_pandas(),
          x="WorkLifeBalance",
          y="Left")
  .add(so.Dot(alpha=.20))
  .add(so.Line(), so.PolyFit(order=1))
)
```





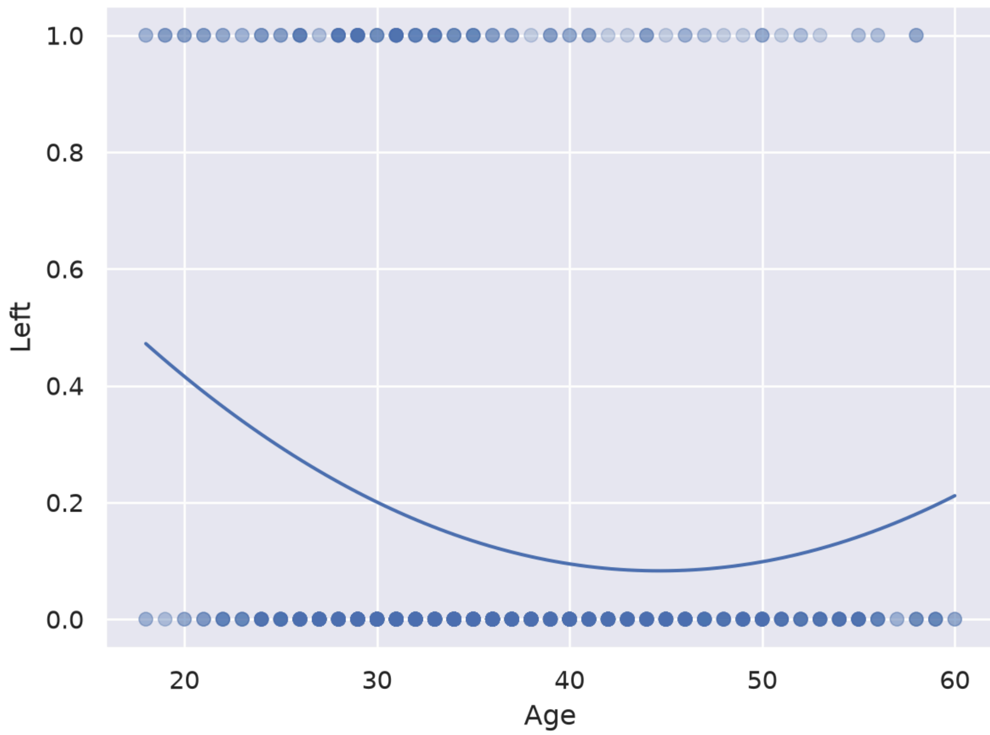
1.0
0.8
0.6
0.4
0.2
0.0
Left
1.0
1.5
2.0
2.5
3.0
3.5
4.0
EnvironmentSatisfaction



These graphs show what we would expect, that as employees report being more satisfied or having better work/life balance, they are less likely to leave.

Age vs Attrition

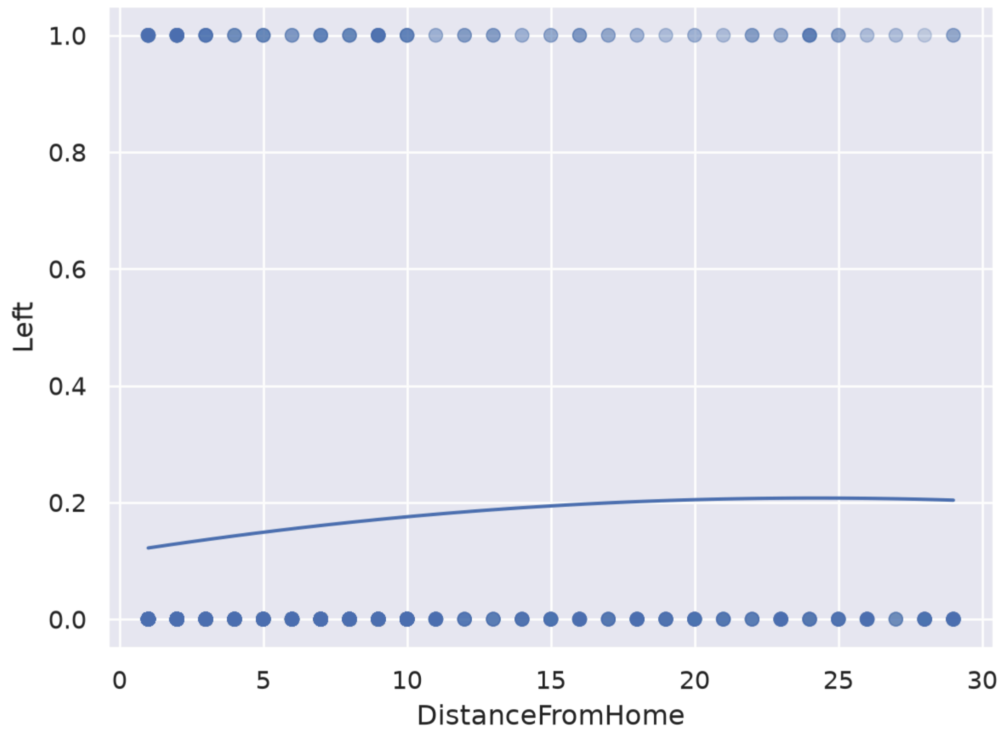
```
(
  so.Plot(employee_data.to_pandas(),
          x="Age",
          y="Left")
  .add(so.Dot(alpha=.15))
  .add(so.Line(), so.PolyFit(order=2))
)
```



We can see that younger employees are generally more likely to leave, but this starts to reverse around age 45. Retirement is likely a factor, but also employees are likely changing jobs for more senior roles.

Commute vs Attrition

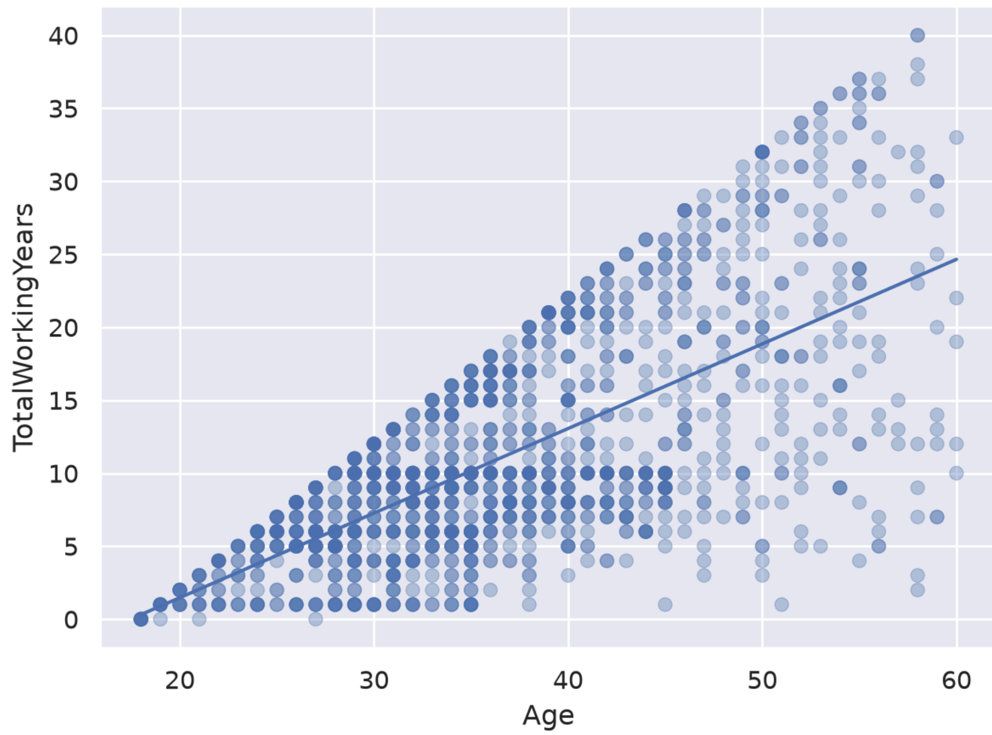
```
(
  so.Plot(employee_data.to_pandas(),
          x="DistanceFromHome",
          y="Left")
  .add(so.Dot(alpha=.15))
  .add(so.Line(), so.PolyFit(order=2))
)
```



There is a fairly weak correlation between how far someone lives from work and if they are likely to leave. It is interesting that the trend appears to level off so it seems like it's only a factor within a certain range.

Age vs Total Working Years

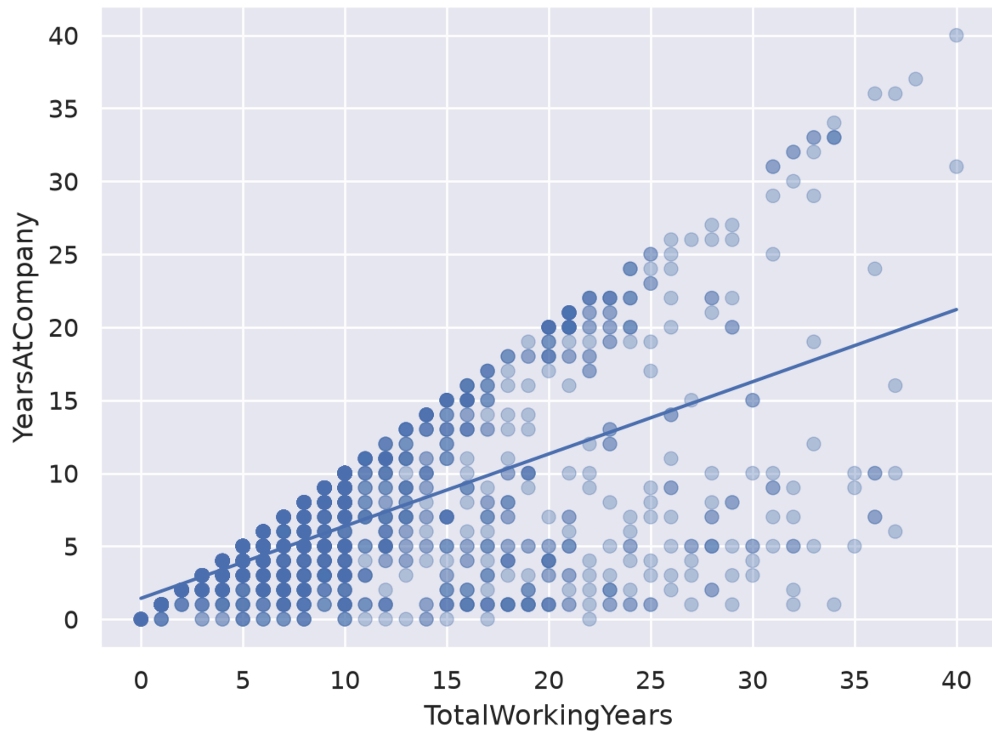
```
(
  so.Plot(employee_data.to_pandas(),
          x="Age",
          y="TotalWorkingYears")
  .add(so.Dot(alpha=.35))
  .add(so.Line(), so.PolyFit(order=1))
)
```



We thought there would be a strong relationship in the data that could cause issues between employee age and total working years. It looks like there is a relationship, but the correlation is not nearly as strong as we thought.

Years at Company vs Total Working Years

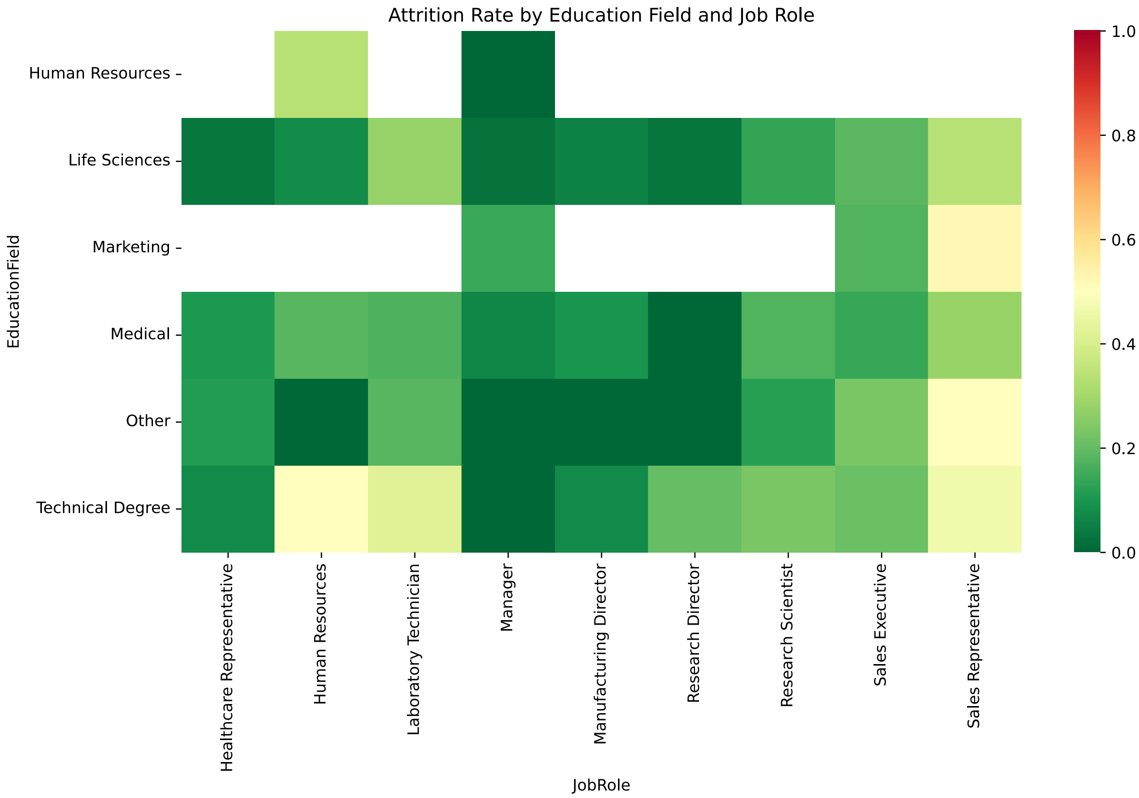
```
(
  so.Plot(employee_data.to_pandas(),
          x="TotalWorkingYears",
          y="YearsAtCompany")
  .add(so.Dot(alpha=.35))
  .add(so.Line(), so.PolyFit(order=1))
)
```



It looks like there is a relationship between years at the company and total working years, but not a very strong one.

Education Field vs Job Role

```
rate = (
  employee_data
  .group_by(["EducationField", "JobRole"])
  .agg(attrition_rate=pl.col("Left").mean())
  .to_pandas()
)
pivot_rate = rate.pivot(
  index="EducationField",
  columns="JobRole",
  values="attrition_rate"
)
plt.figure(figsize=(12,6))
sns.heatmap(pivot_rate, cmap="RdYlGn_r", vmin=0, vmax=1)
plt.title("Attrition Rate by Education Field and Job Role")
plt.show()
```

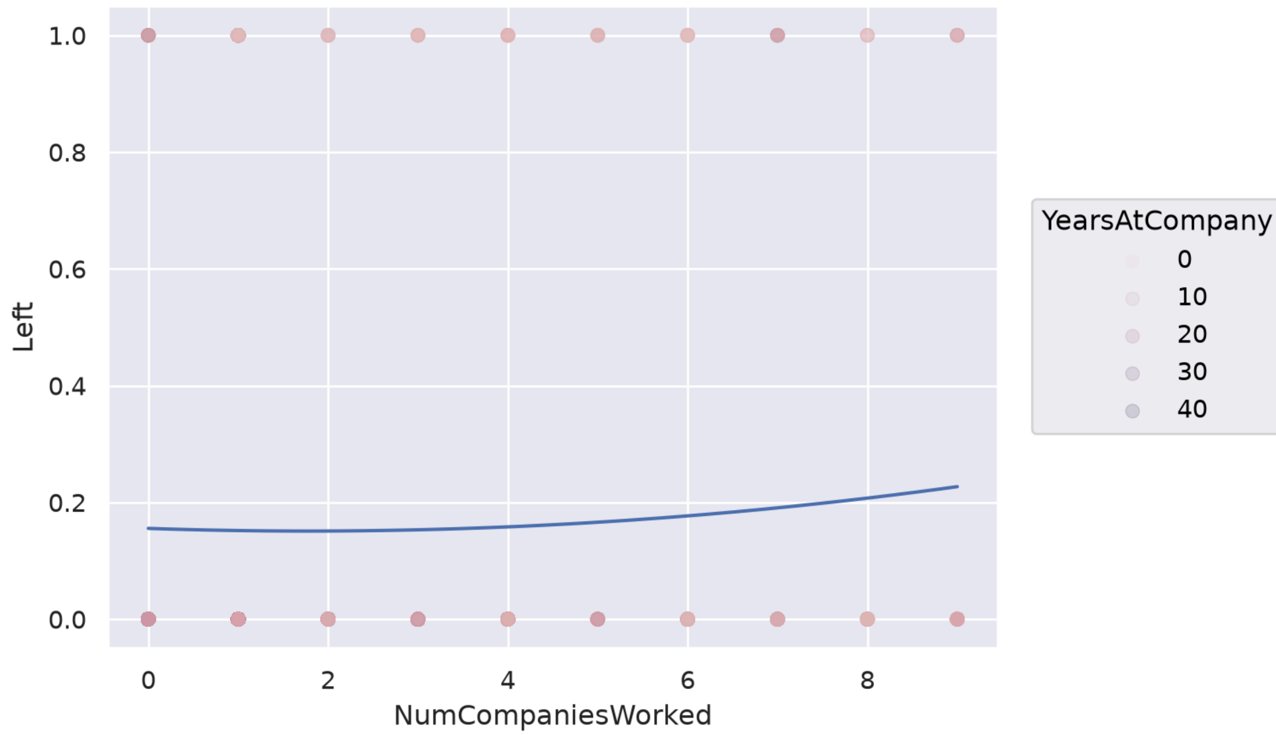


The heatmap reveals a strong structural relationship between education field and job role, indicating that employees are not randomly distributed across roles. Instead, specific education backgrounds concentrate within particular job families. This dependency explains multicollinearity we later observed in the initial logistic regression model.

Attrition varies more meaningfully across job roles than across education fields, suggesting that occupational role is a stronger predictor of turnover than educational background.

Number of Companies and Attrition

```
(
  so.Plot(employee_data.to_pandas(),
          x="NumCompaniesWorked",
          y="Left")
  .add(so.Dot(alpha=.15), color="YearsAtCompany")
  .add(so.Line(), so.PolyFit(order=2))
)
```



This is an interesting graph because it shows that employees that have worked are more companies are more likely to leave. This could be explained by the fact that employees that have worked at more companies are more likely to be looking for new opportunities. It also shows that employees that have been at the company for a long time are less likely to leave, which makes sense.

Reconcile

Sampling Strategy

A stratified sampling approach was used to evenly split test and training data with even attrition and non-attrition employees.

Model Refinements

Several modeling approaches were evaluated, including Ridge, LASSO, and Elastic Net logistic regression models.

Regularization was selected because the dataset contains correlated predictors, particularly among employee tenure, job role, and satisfaction variables. These relationships can create unstable coefficient estimates in traditional logistic regression models.

The final modeling approach optimized both predictive performance and business impact. Rather than maximizing accuracy alone, model selection prioritized expected utility because false positive retention interventions create direct costs.

Hyper-parameter Tuning

```
employee_data = employee_data.to_dummies(
  columns=[
    'BusinessTravel',
    'EducationField',
    'Gender',
    'JobRole',
    'MaritalStatus',
    'OverTime'
  ]
).to_pandas()

# This is a utility function meant to maximize the use of money to retain employees
# It is based on our assumptions:
# * Cost of employee turnover: $40,000
# * Cost of retention intervention: $8,000
# * Retention intervention success rate: 60%
def utility(matrix):
  true_negative, false_positive, false_negative, true_positive = matrix.ravel()

  return (
    true_positive * 16000 +
    false_positive * (-8000) +
    false_negative * 0 +
    true_negative * 0
  )

X = employee_data.drop(columns="Left")
y = employee_data["Left"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Candidate cutoff probabilities and c values
c_values = np.logspace(-2, 2, 15)
thresholds = np.arange(0.20, 0.61, 0.05)

models = {
```

```
  "Ridge": LogisticRegression(l1_ratio=0),
  "Lasso": LogisticRegression(
    l1_ratio=1,
    solver="saga",
    max_iter=5000,
    random_state=42
  ),
  "Elastic": LogisticRegression(
    solver="saga",
    l1_ratio=0.5,
    max_iter=5000,
  )
}

# Setup proper 5-fold cross-validation object
kFolds = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

results = []

# We will try all 3 different models to see which one seems the best
for model_name, base_model in models.items():

  # We are going to tune our cutoff probability and c value at the same time
  best = {
    "C": None,
    "Threshold": None,
    "Utility": -np.inf,
    "Accuracy": 0,
    "Precision": 0,
    "Recall": 0,
    "AUC": 0
  }

  # Loop through each regularization parameter (C)
  for C in c_values:
    for threshold in thresholds:
      fold_utilities = []
      fold_accuracy = []
      fold_precision = []
      fold_recall = []
      fold_auc = []

      # Iterate through the 5 distinct validation folds
```

```
    for train_idx, val_idx in kFolds.split(X_train, y_train):

      # Split using the index arrays generated by StratifiedKFold
      X_train_fold = X_train.iloc[train_idx]
      X_val_fold = X_train.iloc[val_idx]

      y_train_fold = y_train.iloc[train_idx]
      y_val_fold = y_train.iloc[val_idx]

      # grab our frequentist logistic regression model and set C
      model = Pipeline([
        ("scaler", StandardScaler()),
        ("logistic", clone(base_model).set_params(C=C))
      ])

      model.fit(X_train_fold, y_train_fold)

      probabilities = model.predict_proba(X_val_fold)[:,1]

      predictions = (probabilities >= threshold).astype(int)

      cm = confusion_matrix(y_val_fold, predictions)

      fold_utilities.append(utility(cm))
      fold_accuracy.append(accuracy_score(y_val_fold, predictions))
      fold_precision.append(
          precision_score(y_val_fold, predictions, zero_division=0)
      )
      fold_recall.append(
          recall_score(y_val_fold, predictions, zero_division=0)
      )

      # AUC should use probabilities, not binary predictions
      fold_auc.append(
          roc_auc_score(y_val_fold, probabilities)
      )

    avg_utility = np.mean(fold_utilities)
    avg_accuracy = np.mean(fold_accuracy)
    avg_precision = np.mean(fold_precision)
    avg_recall = np.mean(fold_recall)
    avg_auc = np.mean(fold_auc)
```

```python
      if avg_utility > best["Utility"]:
        best["C"] = C
        best["Threshold"] = threshold
        best["Utility"] = avg_utility
        best["Accuracy"] = avg_accuracy
        best["Precision"] = avg_precision
        best["Recall"] = avg_recall
        best["AUC"] = avg_auc

      # It's very possible we tie, if so go by accuracy
      elif (
          avg_utility == best["Utility"]
          and avg_accuracy > best["Accuracy"]
      ):
        best["C"] = C
        best["Threshold"] = threshold
        best["Utility"] = avg_utility
        best["Accuracy"] = avg_accuracy
        best["Precision"] = avg_precision
        best["Recall"] = avg_recall
        best["AUC"] = avg_auc

  # Save the best settings for this model
  results.append({
    "Model": model_name,
    **best
  })

# Identify the best cutoff probability, best C, and best model
results = pd.DataFrame(results)

results.sort_values(
    "Utility",
    ascending=False
).head()
```

	Model	C	Threshold	Utility	Accuracy	Precision	Recall	AUC
0	Ridge	0.071969	0.35	260800.0	0.885204	0.671458	0.568421	0.842234
2	Elastic	0.517947	0.40	260800.0	0.889452	0.701604	0.542105	0.841832
1	Lasso	1.000000	0.40	257600.0	0.887753	0.690919	0.542105	0.841591

Elastic Net and Ridge achieved the highest cross-validated business utility, each producing an expected utility of $260,800. Elastic Net was selected as the final model because it achieved comparable utility while providing slightly better accuracy and precision.

Precision was particularly important because false positives represent unnecessary retention interventions costing $8,000 each. The selected model therefore prioritizes identifying employees at meaningful risk while avoiding excessive intervention costs.

The similarity in performance suggests that the employee attrition dataset contains a relatively stable set of predictors with limited multicollinearity. Consequently, the choice of regularization method had only a modest impact on predictive performance. Our removal from the data of some predictors where we saw or predicted multicollinearity may have also contributed.

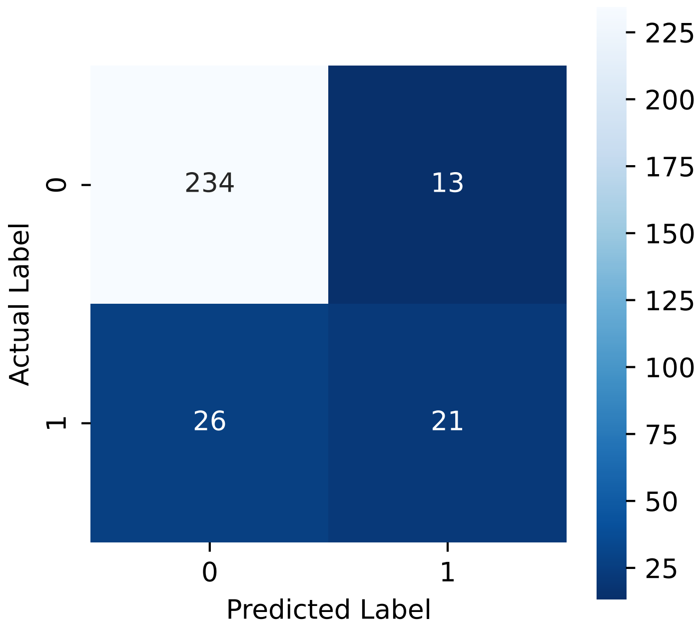
```
final_c = 0.517947
final_probability = 0.40

final_model = Pipeline([
    ("scaler", StandardScaler()),
    ("classification", LogisticRegression(
        solver="saga",
        l1_ratio=0.5,
        C=final_c,
        max_iter=5000,
        random_state=42
    ))
])

final_model.fit(X_train, y_train)

p_test = final_model.predict_proba(X_test)[:, 1]
y_pred = (p_test >= final_probability).astype(int)
final_accuracy = accuracy_score(y_test, y_pred)

final_accuracy

cm = confusion_matrix(y_test, y_pred)
print(utility(cm))

employee_data = employee_data.rename(columns={
    "BusinessTravel_Non-Travel": "BusinessTravel_NonTravel",
    "EducationField_Human Resources": "EducationField_HumanResources",
    "EducationField_Life Sciences": "EducationField_LifeSciences",
    "EducationField_Technical Degree": "EducationField_TechnicalDegree",
    "JobRole_Healthcare Representative": "JobRole_HealthcareRepresentative",
```

```python
    "JobRole_Human Resources": "JobRole_HumanResources",
    "JobRole_Laboratory Technician": "JobRole_LaboratoryTechnician",
    "JobRole_Manufacturing Director": "JobRole_ManufacturingDirector",
    "JobRole_Research Director": "JobRole_ResearchDirector",
    "JobRole_Research Scientist": "JobRole_ResearchScientist",
    "JobRole_Sales Executive": "JobRole_SalesExecutive",
    "JobRole_Sales Representative": "JobRole_SalesRepresentative",
})
```

```
232000
```

When evaluated on the held-out test set, the final model produced an expected utility of $232,000. This demonstrates that the model generated positive expected business value on unseen employees rather than only during cross-validation.

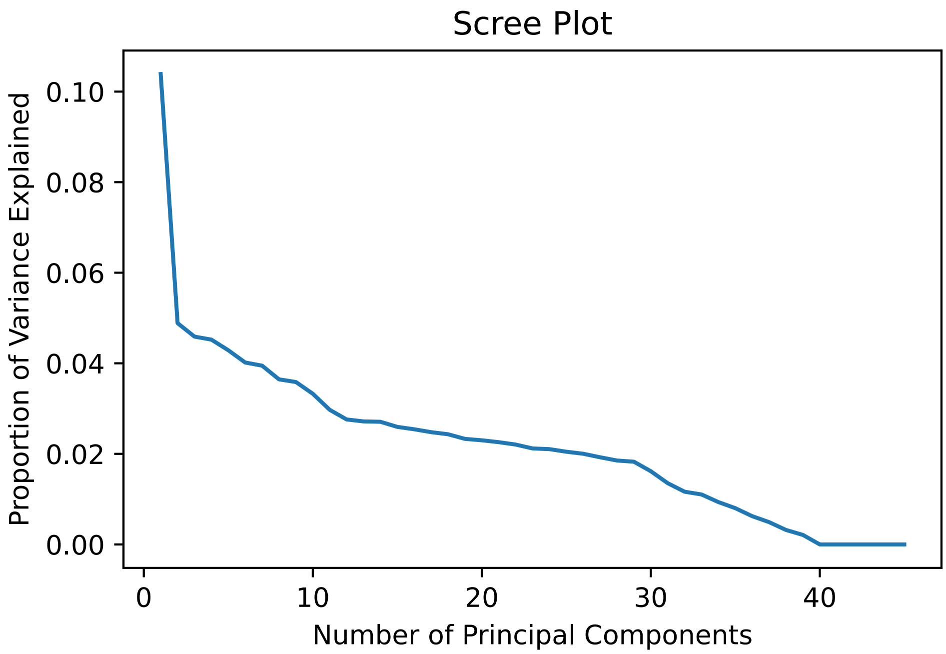
```python
# Confusion matrix with labels
plt.figure(figsize = (4, 4))
sns.heatmap(
    cm,
    annot = True,
    fmt = ".0f",
    square = True,
    cmap = 'Blues_r'
)
plt.ylabel('Actual Label')
plt.xlabel('Predicted Label')
plt.show()
```



PCA Analysis

```
# Try PCA
from sklearn.decomposition import PCA

# Split data into train and test
X_train_nostrat, X_test_nostrat = train_test_split(X, test_size = 0.2,
random_state = 42)

# Create a pipeline
pca_pipe = Pipeline([
    ('feature_engineering', StandardScaler()),
    ('pca', PCA(n_components = X_train_nostrat.shape[1])),
])

# Fit PCA
pca_fit = pca_pipe.fit(X_train_nostrat)

# Extract explained variance ratio
pca = pca_fit.named_steps['pca']
pca_scores = pca.explained_variance_ratio_

# Create scree plot
sns.lineplot(
    x = np.arange(1, X_train_nostrat.shape[1] + 1),
```

```
    y = pca_scores
)
plt.xlabel('Number of Principal Components')
plt.ylabel('Proportion of Variance Explained')
plt.title('Scree Plot')
plt.show()

pca_predictors = X.columns

# Extract loadings and rename
pca_loadings = (pl.DataFrame(pca.components_)
       .select(['column_0',  'column_1',  'column_2',  'column_3',  'column_4',
'column_5',  'column_6',  'column_7',  'column_8',  'column_9',  'column_10',
'column_11'])
    .rename({
        'column_0': 'PC1',
        'column_1': 'PC2',
        'column_2': 'PC3',
        'column_3': 'PC4',
        'column_4': 'PC5',
        'column_5': 'PC6',
        'column_6': 'PC7',
        'column_7': 'PC8',
        'column_8': 'PC9',
        'column_9': 'PC10',
        'column_10': 'PC11',
        'column_11': 'PC12',
    })
).with_columns(pl.Series('predictors', pca_predictors))
pca_loadings.select(['predictors',     'PC1']).sort('PC1',     descending     =
True).head(10)
```



predictors	PC1
str	f64
"MonthlyRate"	0.626475
"MaritalStatus_Single"	0.290598
"Age"	0.280073
"EducationField_Human Resources"	0.237846
"StockOptionLevel"	0.197103
"NumCompaniesWorked"	0.185223
"DistanceFromHome"	0.162584
"EducationField_Medical"	0.119586
"BusinessTravel_Travel_Rarely"	0.108766
"OverTime_No"	0.107833

```
# Sort the loadings for PC2
pca_loadings.select(['predictors', 'PC2']).sort('PC2', descending =
True).head(10)
```

predictors	PC2
str	f64
"EnvironmentSatisfaction"	0.375616
"JobLevel"	0.307723
"YearsWithCurrManager"	0.279728
"YearsInCurrentRole"	0.279167
"JobRole_Human Resources"	0.216668
"JobRole_Sales Representative"	0.210629
"YearsSinceLastPromotion"	0.145894
"EducationField_Human Resources"	0.138389
"JobRole_Manufacturing Director"	0.117714
"EducationField_Technical Degree"	0.100136

```
# Sort the loadings for PC3
pca_loadings.select(['predictors', 'PC3']).sort('PC3', descending =
True).head(10)
```

predictors	PC3
str	f64
"YearsWithCurrManager"	0.375233
"YearsInCurrentRole"	0.37448
"EducationField_Medical"	0.303656
"Gender_Male"	0.230056
"EducationField_Human Resources"	0.217253
"Gender_Female"	0.202156
"YearsSinceLastPromotion"	0.195706
"TrainingTimesLastYear"	0.128449
"JobRole_Manager"	0.097202
"EducationField_Other"	0.063596

```
# Sort the loadings for PC4
pca_loadings.select(['predictors', 'PC4']).sort('PC4', descending =
True).head(10)
```

predictors	PC4
str	f64
"DistanceFromHome"	0.434621
"YearsWithCurrManager"	0.430675
"YearsInCurrentRole"	0.429811
"BusinessTravel_Travel_Rarely"	0.40591
"YearsSinceLastPromotion"	0.224622
"TrainingTimesLastYear"	0.147428
"BusinessTravel_Travel_Frequently"	0.122501
"Gender_Male"	0.072431
"JobRole_Research Director"	0.064776
"JobSatisfaction"	0.044428

PCA reduced the dimensionality of the dataset, with diminishing variance contributions after approximately 12 components. However, the resulting components did not correspond to clearly interpretable employee characteristics because individual components combined demographic, job, compensation, and tenure variables.

Since Elastic Net logistic regression already addresses correlated predictors while preserving interpretable features, PCA-based modeling was not selected for the final approach.

Fit

Final Model

```
final_model_full = Pipeline([
    ("scaler", StandardScaler()),
    ("classification", LogisticRegression(
        solver="saga",
        l1_ratio=0.5,
        C=final_c,
        max_iter=5000,
        random_state=42
    ))
])

final_model_full.fit(X, y)

coef_df = (
  pd.DataFrame({
```

```
    "Predictor": X.columns,
    "Coefficient": final_model_full.named_steps["classification"].coef_[0]
  })
  .assign(
    Odds_Ratio=lambda x: np.exp(x["Coefficient"])
  )
  .sort_values(
    "Coefficient",
    ascending=False
  )
)
```

Bootstrap Confidence Intervals

```
from sklearn.utils import resample
from sklearn.linear_model import LogisticRegression

n_iterations = 1000
boot_coefs = []

X_train_clean = X_train.drop(columns=['Intercept'], errors='ignore')

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_clean)

for i in range(n_iterations):
    # Resample training data
    X_boot, y_boot = resample(X_train_scaled, y_train, random_state=i)

    # Refit model with the fixed optimal hyperparameter
      boot_model = LogisticRegression(l1_ratio=0.5, solver='saga', C=final_c,
max_iter=5000, random_state=42)
    boot_model.fit(X_boot, y_boot)
    boot_coefs.append(boot_model.coef_.flatten())

boot_coefs = np.array(boot_coefs)

# Calculate 95% Percentile Confidence Intervals
lower_bounds = np.percentile(boot_coefs, 2.5, axis=0)
upper_bounds = np.percentile(boot_coefs, 97.5, axis=0)

ci_df = pd.DataFrame({
    'Feature': X_train_clean.columns,
```

```
    'Optimal_Coef': boot_model.coef_.flatten(),
    '2.5%': lower_bounds,
    '97.5%': upper_bounds
})
print(ci_df)
```

```
                              Feature  Optimal_Coef      2.5%     97.5%
0                                 Age     -0.363739 -0.728848 -0.082505
1           BusinessTravel_Non-Travel     -0.265356 -0.587555 -0.105176
2    BusinessTravel_Travel_Frequently      0.424046  0.181975  0.549378
3        BusinessTravel_Travel_Rarely      0.000000  0.000000  0.000000
4                    DistanceFromHome      0.348508  0.188686  0.575625
5                           Education      0.035551 -0.166596  0.202203
6      EducationField_Human Resources      0.103594 -0.013504  0.344272
7        EducationField_Life Sciences      0.000000 -0.151258  0.132224
8            EducationField_Marketing      0.189447 -0.080347  0.281892
9              EducationField_Medical     -0.049055 -0.297073  0.034224
10               EducationField_Other     -0.072893 -0.485287  0.000255
11    EducationField_Technical Degree      0.161404  0.000000  0.349063
12            EnvironmentSatisfaction     -0.442040 -0.703227 -0.279672
13                      Gender_Female     -0.103835 -0.175213  0.019587
14                        Gender_Male      0.103835 -0.019587  0.175213
15                         HourlyRate      0.141777 -0.215775  0.167301
16                     JobInvolvement     -0.343945 -0.578676 -0.189995
17                           JobLevel      0.273450 -0.208473  0.705606
18  JobRole_Healthcare Representative     -0.203936 -0.606919  0.000000
19           JobRole_Human Resources      0.081555 -0.106171  0.257042
20      JobRole_Laboratory Technician      0.396873  0.177909  0.613267
21                    JobRole_Manager     -0.200968 -0.417214  0.156988
22     JobRole_Manufacturing Director     -0.018044 -0.504230  0.011228
23          JobRole_Research Director     -0.395224 -0.930583 -0.107048
24         JobRole_Research Scientist      0.000000 -0.345913  0.023468
25            JobRole_Sales Executive      0.174694  0.000000  0.412139
26       JobRole_Sales Representative      0.320465  0.157587  0.509597
27                    JobSatisfaction     -0.300835 -0.644665 -0.227989
28             MaritalStatus_Divorced     -0.118561 -0.286365  0.000000
29              MaritalStatus_Married      0.000000 -0.212382  0.000000
30               MaritalStatus_Single      0.458455  0.037700  0.598234
31                        MonthlyRate     -0.023955 -0.171303  0.213903
32                 NumCompaniesWorked      0.582944  0.272875  0.675616
33                        OverTime_No     -0.451459 -0.540525 -0.331858
34                       OverTime_Yes      0.451459  0.331858  0.540525
```

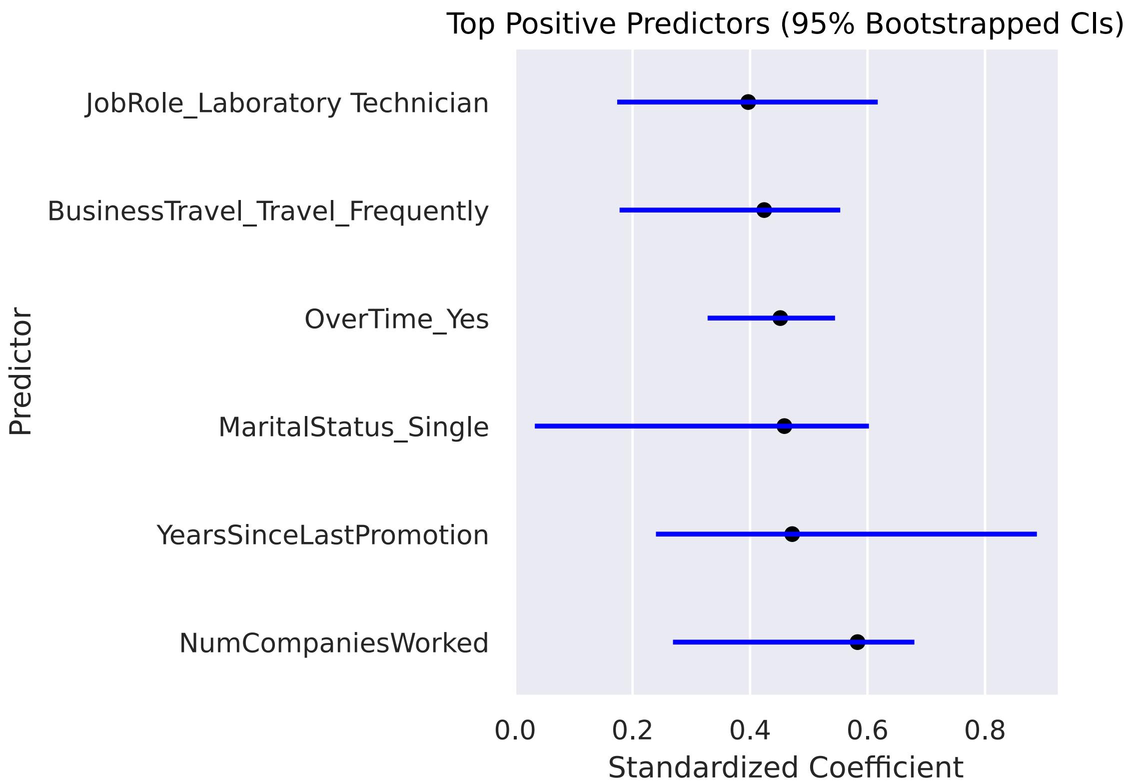
```
35                   PercentSalaryHike     -0.039945 -0.274104  0.113741
36           RelationshipSatisfaction     -0.387828 -0.520161 -0.068507
37                    StockOptionLevel     -0.023451 -0.642062  0.033432
38                   TotalWorkingYears     -0.707473 -1.012441 -0.026431
39               TrainingTimesLastYear     -0.186286 -0.388010  0.007209
40                     WorkLifeBalance     -0.303536 -0.512531 -0.078343
41                      YearsAtCompany      0.133094 -0.392533  0.591344
42                  YearsInCurrentRole     -0.313912 -0.705821 -0.032934
43             YearsSinceLastPromotion      0.471753  0.243899  0.884166
44                YearsWithCurrManager     -0.423168 -0.755593  0.000000
```

Predictors with stable positive association

Predictor	Coefficient	95% CI
NumCompaniesWorked	0.583	[0.273, 0.676]
YearsSinceLastPromotion	0.472	[0.244, 0.884]
OverTime_Yes	0.451	[0.332, 0.541]
BusinessTravel_Frequently	0.424	[0.182, 0.549]
JobRole_LaboratoryTechnician	0.397	[0.178, 0.613]
DistanceFromHome	0.349	[0.189, 0.576]

```python
ci_df_sorted = ci_df.sort_values("Optimal_Coef", ascending=True)

top_pos_coef = ci_df_sorted.tail(6)
top_neg_coef = ci_df_sorted.head(6)
zero_line_df = pd.DataFrame({"x": [0, 0]})

def plot_tie_fighter(df, title):
    return (
        so.Plot(df, x="Optimal_Coef", y="Feature", xmin="2.5%", xmax="97.5%")
        .add(so.Range(linewidth=2, color="blue"))  # Wings (95% CI)
        .add(so.Dot(color="black"))     # Cockpit (Point Estimate)
         .add(so.Line(linestyle="--", color="red"), x=zero_line_df['x'])      #
Zero Reference Line
        .label(
            x="Standardized Coefficient",
            y="Predictor",
            title=title
        )
    )
```

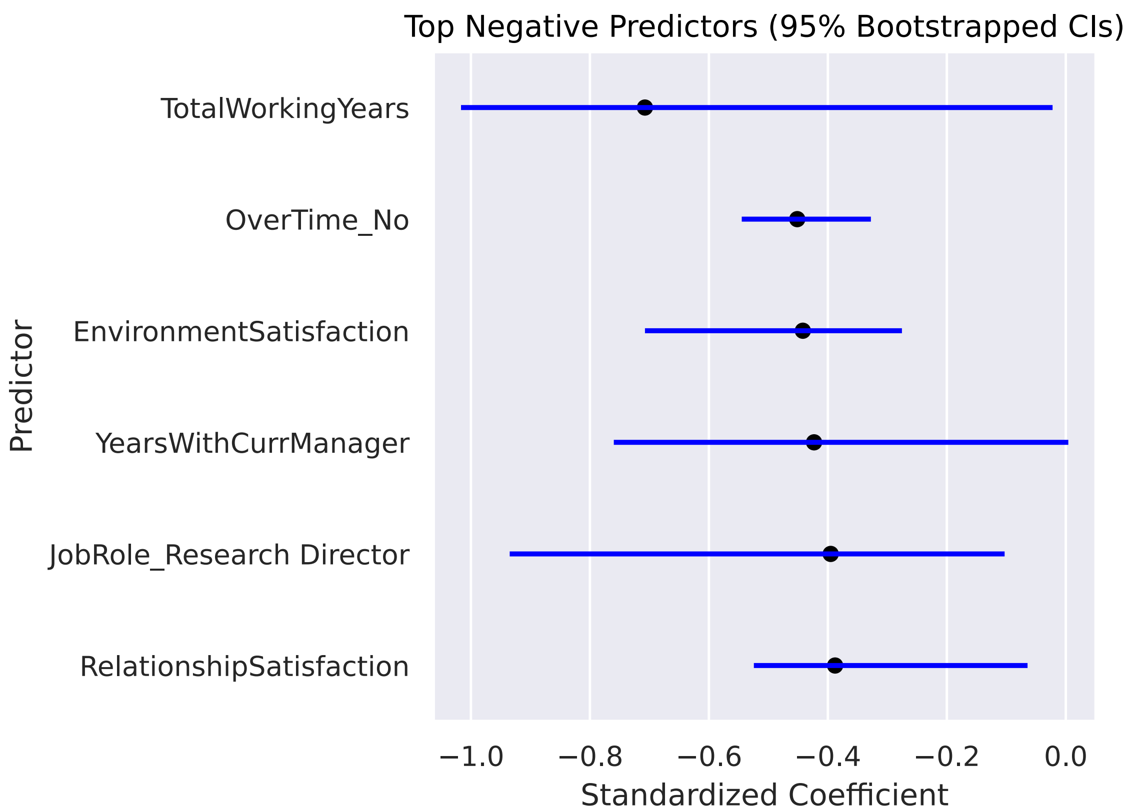
```
plot_tie_fighter(top_pos_coef, "Top Positive Predictors (95% Bootstrapped CIs)").show()
```



Predictors with stable negative association

Predictor	Coefficient	95% CI
TotalWorkingYears	−0.707	[−1.012, −0.026]
EnvironmentSatisfaction	−0.442	[−0.703, −0.280]
OverTime_No	−0.451	[−0.541, −0.332]
YearsWithCurrManager	−0.423	[−0.756, 0.000]
JobSatisfaction	−0.301	[−0.645, −0.228]

```
plot_tie_fighter(top_neg_coef, "Top Negative Predictors (95% Bootstrapped CIs)").show()
```



Interpretation of Final Model Coefficients

The final Elastic Net logistic regression model was examined using bootstrap confidence intervals generated from 1,000 resampled training datasets. Predictors with confidence intervals that remained consistently above or below zero were considered more stable contributors to attrition prediction.

The strongest positive predictors were related to employee mobility and work conditions. Employees who frequently traveled for business, worked overtime, had more previous employers, and had longer periods since their last promotion showed increased attrition risk.

Conversely, higher job satisfaction, environment satisfaction, job involvement, and total working years were associated with lower attrition risk.

Predictor	Coefficient	95% Bootstrap Interval
NumCompaniesWorked	0.583	(0.273, 0.676)
OverTime_Yes	0.451	(0.332, 0.541)
BusinessTravel_Frequently	0.424	(0.182, 0.549)
DistanceFromHome	0.349	(0.189, 0.576)
TotalWorkingYears	−0.707	(−1.012, −0.026)

Predictor	Coefficient	95% Bootstrap Interval
EnvironmentSatisfaction	−0.442	(−0.703, −0.280)
JobSatisfaction	−0.301	(−0.645, −0.228)

Model Diagnostics

Because the final model is a classification model rather than an ordinary least squares regression model, traditional residual assumptions are not applicable. Model quality was evaluated through cross-validation, held-out test performance, ROC-AUC, and business utility.

Potential multicollinearity was addressed through preprocessing decisions and regularization. Elastic Net was appropriate because it reduces the influence of correlated predictors while retaining groups of related variables when they provide predictive value.

Limitations

Several limitations should be considered:

- The dataset is synthetic and may not represent real employee behavior or organizational conditions.
- The model identifies associations with attrition but does not establish causal relationships.
- The dataset represents a single organization, so performance may not generalize to other companies.
- Retention intervention costs and success rates were estimated assumptions rather than measured business outcomes.

These limitations mean the model should be used as a decision-support tool rather than an automatic decision system.

Predict

The selected Elastic Net model demonstrated measurable business value on the held-out test set, generating an expected utility of $232,000 under the proposed retention strategy. This indicates that targeting interventions using model predictions provides substantially greater expected value than applying no targeted retention strategy.

The final Elastic Net model was refit using the full dataset after model selection. Because all continuous predictors were standardized prior to fitting, coefficient magnitudes represent the effect of a one-standard-deviation increase in each predictor while holding the remaining variables constant. Larger absolute coefficients therefore indicate predictors with stronger influence on the model.

The bootstrap analysis identified several stable predictors of attrition. Refitting the selected Elastic Net model on the complete dataset produced similar coefficient rankings, providing additional confidence that these relationships are consistent across the data.

The largest positive standardized coefficients in the refit model were:

```
coef_df.head(10)
```

	Predictor	Coefficient	Odds_Ratio
43	YearsSinceLastPromotion	0.542702	1.720649
32	NumCompaniesWorked	0.445739	1.561644
34	OverTime_Yes	0.426897	1.532495
41	YearsAtCompany	0.407288	1.502736
30	MaritalStatus_Single	0.372748	1.451718
4	DistanceFromHome	0.345652	1.412911
20	JobRole_Laboratory Technician	0.337355	1.401236
2	BusinessTravel_Travel_Frequently	0.331192	1.392627
26	JobRole_Sales Representative	0.308366	1.361199
11	EducationField_Technical Degree	0.242302	1.274180

The largest positive standardized coefficients were associated with employees who had gone longer since their last promotion, worked for more previous employers, regularly worked overtime, had longer tenure with the company, were single, lived farther from work, or occupied certain job roles. Frequent business travel was also associated with increased attrition risk, although its estimated effect was somewhat smaller than several of these other predictors.

Conversely, several predictors were associated with reduced attrition risk:

```
coef_df.tail(10)
```

	Predictor	Coefficient	Odds_Ratio
0	Age	−0.279078	0.756481
1	BusinessTravel_Non-Travel	−0.284759	0.752196
23	JobRole_Research Director	−0.293820	0.745411
16	JobInvolvement	−0.367697	0.692327
44	YearsWithCurrManager	−0.401428	0.669364
38	TotalWorkingYears	−0.405297	0.666778
33	OverTime_No	−0.426897	0.652531
27	JobSatisfaction	−0.435684	0.646822
12	EnvironmentSatisfaction	−0.447595	0.639164
42	YearsInCurrentRole	−0.468833	0.625732

These results suggest that employee stability is associated with satisfaction measures and longer-term relationships within the organization.

The most reliable findings are those that both have relatively large standardized coefficients and bootstrap confidence intervals that do not cross zero. These predictors are most consistently associated with employee attrition across repeated samples of the data. These predictors include:

Predictor	Result
YearsSinceLastPromotion	strongest positive effect and stable CI
NumCompaniesWorked	strong positive effect and stable CI
OverTime	strong positive effect and stable CI
BusinessTravel (Frequent)	moderate positive effect and stable CI
DistanceFromHome	moderate positive effect and stable CI
EnvironmentSatisfaction	strong negative effect and stable CI
JobSatisfaction	negative effect and stable CI
TotalWorkingYears	negative effect and stable CI

Business Implications

The strongest findings from the final model were:

- Employees working overtime had approximately 53% higher estimated odds of attrition.
- Employees frequently traveling for business had approximately 39% higher estimated odds of attrition.
- Lower satisfaction scores were consistently associated with increased attrition risk.
- Longer periods without promotion were associated with increased attrition risk.
- Employees with more previous employers showed increased attrition risk.

Recommendations

Based on these findings, retention efforts should focus on:

1. Monitoring employees with sustained overtime requirements.
2. Reviewing workload and travel expectations for frequently traveling employees.
3. Providing career development opportunities for employees experiencing stalled progression.
4. Using satisfaction measures as early indicators of disengagement.

Future Work

Future improvements could include:

1. Validating the model using real organizational attrition data.
2. Adding additional employee engagement and compensation variables.
3. Measuring whether targeted interventions actually reduce attrition after identification.

Conclusion

This project developed a predictive employee attrition model using Elastic Net logistic regression. The final model balanced predictive performance with business value by optimizing an intervention-based utility function rather than accuracy alone.

The selected model achieved a cross-validated utility of $260,800 and generated $232,000 in expected utility on unseen test data. These results indicate that predictive modeling can provide measurable value by helping organizations focus retention resources on employees most likely to leave.

The final Elastic Net model produced positive expected business value while identifying several consistent drivers of attrition. Employees with longer periods since promotion, more previous employers, overtime work, and frequent business travel exhibited higher attrition risk, while higher job satisfaction, environment satisfaction, and greater overall work experience were associated with lower attrition risk. These findings suggest that retention efforts should prioritize employees exhibiting multiple high-risk characteristics rather than relying on any single predictor.